Ex.No.-1.1 – Design and Draft the Given 2D Sketches in Modelling Software

AIM:

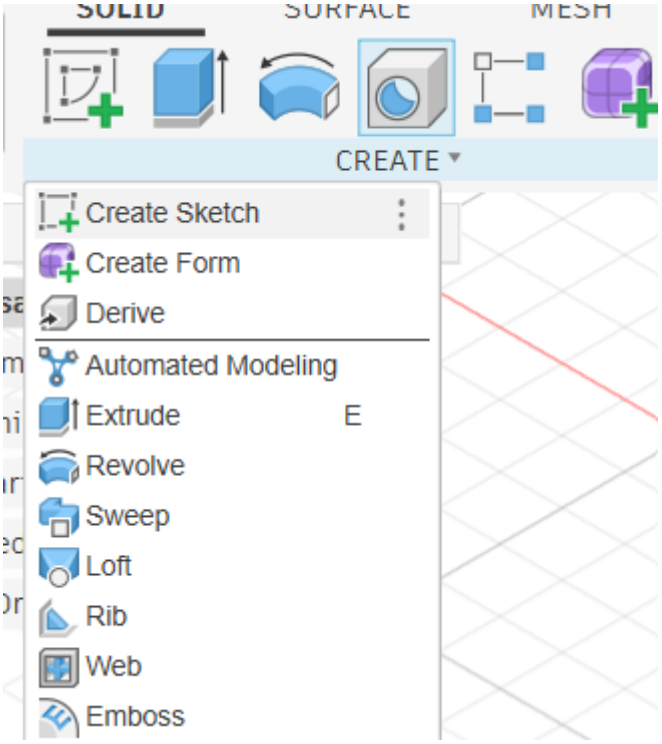
To create a 2D sketch with the specified dimensions and prepare the technical drawing using Autodesk Fusion 360 software.

REQUIREMENTS:

Autodesk Fusion 360 software
Windows 10 operating system
16 GB RAM
Integrated graphics with 2 GB or more recommended
Stable internet connection

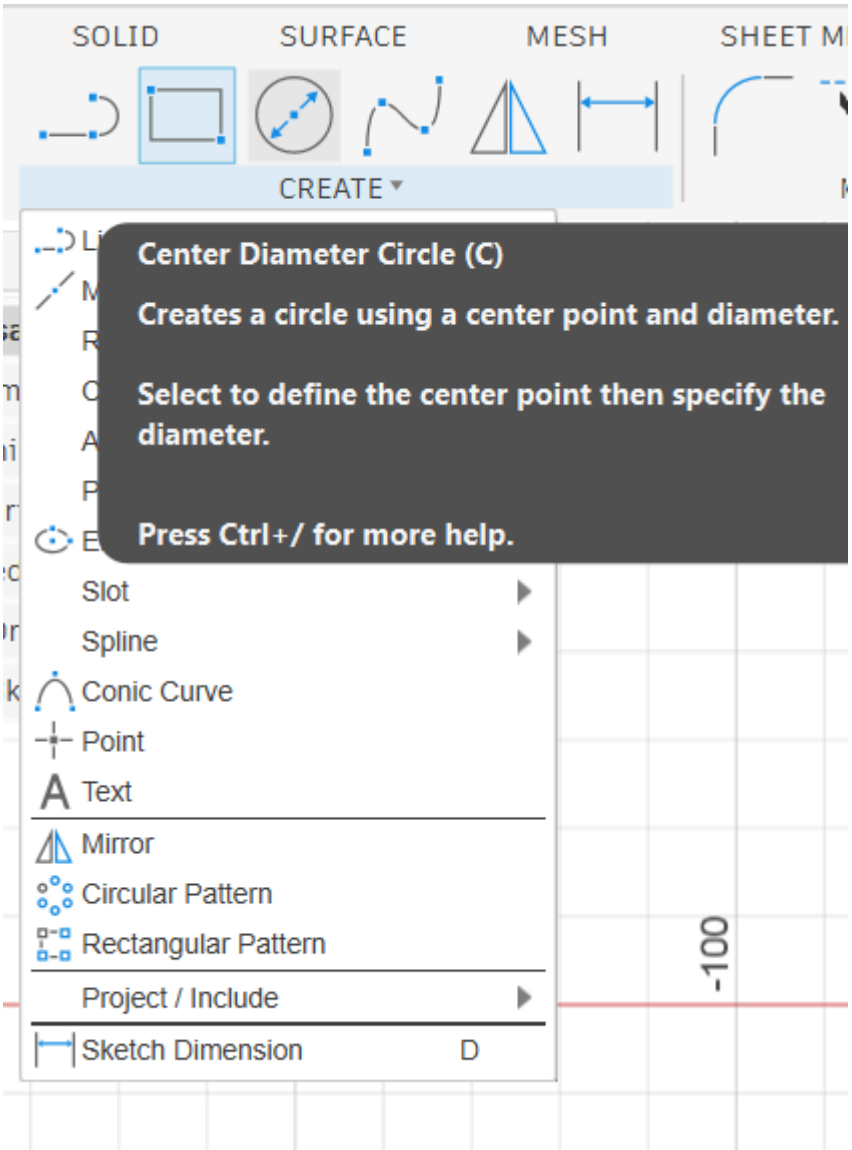
PROCEDURE:

1. Launch Autodesk Fusion 360 and create a new design file. Select the Create Sketch command to begin the drawing.



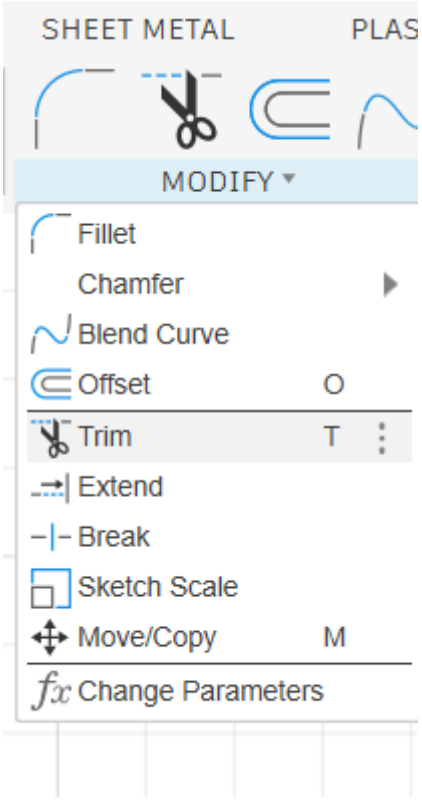
2. Choose a suitable reference plane, such as the XY plane, according to the given drawing.

3. Construct the basic geometry of the sketch using appropriate tools such as Line, Circle, Rectangle, Arc, and Polygon.



4. Apply the required dimensions to the sketch using the Sketch Dimension tool to maintain the specified measurements.

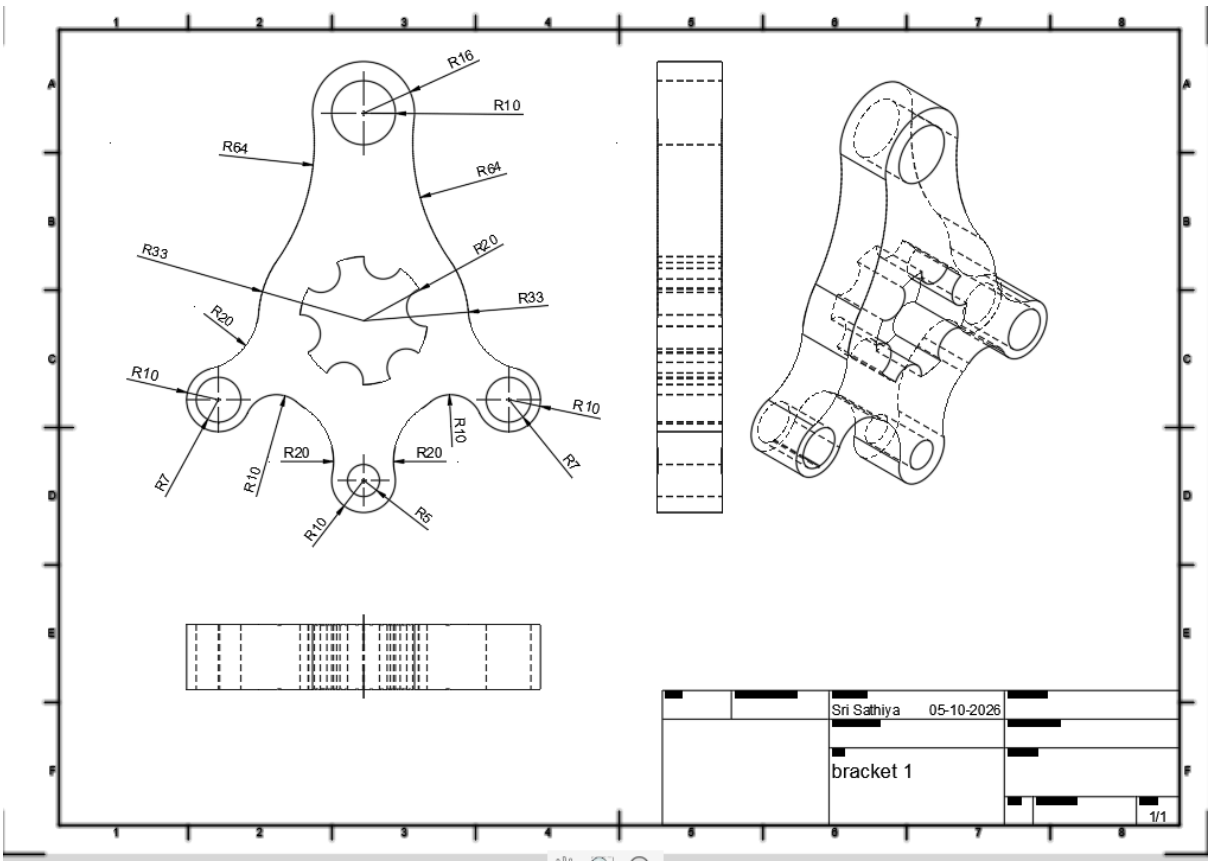
5. Modify the sketch wherever necessary by using editing commands such as Trim, Extend,



Mirror, and Move.
6. Check the shape and measurements of the completed sketch using the available inspection and dimensioning tools.
7. Click Finish Sketch to complete the 2D sketch and save the design.
8. Open the Drawing workspace and create a drawing from the completed sketch or model.
9. Select the ISO standard and adjust the drawing scale and view orientation to display the sketch clearly.
10. Save the final drawing for documentation and submission.
SKETCHES
Attach the given sketch images here.

OUTPUT:

The completed 2D sketch and its technical drawing created using Autodesk Fusion 360.



RESULT:

Thus, the given 2D sketch was designed with the required dimensions and drafted successfully using Autodesk Fusion 360.

